

SRW3 Phase 1F — Authenticated Execution-Effect Binding and Commitment-Lineage Integration

****Status:** PHASE 1F EXECUTED — the Level-3 research question is answered with a qualified, boundary-honest result: the binding machinery is PROVED, the execution-grounded closure is DEMONSTRATED, and the authenticity boundary of the abstract layer is stated, not assumed away.** This report was produced from the phase1f branch derived from the Phase 1E closure commit (8b26cb749... = closure; scientific anchor 0d826c0f1...), on a rebuilt toolchain (K v7.1.337, Z3 4.13.3-1, clang/LLVM-15 15.0.6-4+b1, KEVM v1.0.921 source, blockchain-k-plugin @ 207ae512, crypto shim rebuilt from source with the keccak-multiblock fix). The frozen layers were not modified: `git diff` over `k/`, `python-gen/` and `phase1e/` against the Phase 1E closure state is empty, and every Phase 1F extension is additive (`linCtxP` / `linRecE` / `execCtx` discipline).

1. Research question and answer

Question. Can the SRW3 commitment gate bind its security decision to authenticated evidence of EXECUTION-DERIVED effects — so that the lineage records what a concrete execution actually produced (``PresentedEffects == TrueEffects(execution)``, Level 3) — rather than only (Level 1) a self-consistent presentation or (Level 2) an anchored state representation?

****Answer:** YES at Level 3 as a BINDING result, under explicitly stated assumptions, with the three-level separation preserved — and the authenticity boundary stated honestly.** The gate was extended (additively) so that acceptance requires, on top of the full Phase 1D-R1 + 1E certificate:

1. the declared effects to be byte-canonically EQUAL to the write projection of a digest-bound, order-preserving effect trace (`invalid-effdecl` — the Level-3 core check; effect equality becomes DECIDABLE rather than post-relative);
2. the effect trace to be bound into the record digest chain and coherent with an execution WITNESS (`invalid-efftrace`, `invalid-witness`);
3. the execution identity to be exactly the domain-separated recomputation over (parent state root, payload, chain config, environment) (`invalid-execid`, `invalid-execonfig`) — replaying the same payload against a different parent root, caller, or fork spec yields a different

identity;

4. every presented read observation to be REPLAY-CONSISTENT and the trace's writes to compose exactly to the presented post (`invalid-effbind`) — this mechanizes the oracle-staleness class (false read observations are rejected even when every digest is coherently re-forged);
5. the whole presentation to respect explicit fragment bounds (`invalid-execbounds` — a model refuse, never silent).

What this result IS NOT, stated plainly: at the ABSTRACT layer the trace is a presented input (A-F3). A producer able to forge ALL presented artifacts consistently (trace + record + witness, co-signed) produces an internally consistent Level-3 record — the same authorized-dishonest-producer boundary as every prior phase, now WITH the boundary explicitly mechanized as a negative-control-adjacent disclosure (mutation b02) rather than hidden. Closing that boundary requires an execution-derived trace source: Phase 1F DEMONSTRATES the closure at the instrumented-execution layer (KEVM), where the trace is built by the semantics from real opcode execution and never passes through producer hands; verifier re-execution remains REQUIRES CLIENT/PROTOCOL SUPPORT.

2. Starting state and baseline regression

The handoff discipline mandated re-running the permanent suites before any Phase 1F work (`transcripts/part0_baseline_k_composition.txt`):

Suite	Result
L7 permanent adversarial suite (Python)	10/10 PASS
Legacy hostile-mutation harness (21 classes)	21/21 rejected
Comparison audit	33/33 probes, zero deviations
Authorized-dishonest producer	REJECTED at L7-STATE
Phase 1E Python suite + mutations	24/24 + 12/12
K composition suite (fresh LLVM re-kompile)	pos=valid; neg1=neg2=neg3=invalid-state-composition

Environment note, recorded honestly: this session began from a sandbox reset. The remote `phase1e` branch was re-fetched from GitHub (the local worktree and the local worklog had no Phase 1E state — a discrepancy with the session-start expectation, recorded in the worklog); the toolchain was rebuilt from the pinned recipe; the anchors `0d826c0f` (scientific) and `8b26cb74` (closure) were verified on the fetched branch before `phase1f` was derived.

3. The minimal execution-effect model (FROZEN)

phase1f/semantics/MODEL.md is the normative specification. Design points:

- **Vocabulary (fragment).** Read(1) / Write(2) / Call(3) / Return(4) events exEvt(op, app, a1, a2) sourced from SLOAD / SSTORE / CALL / RETURN. LOGn, CREATE/CREATE2, CALLCODE, DELEGATECALL, STATICCALL, SELFDESTRUCT, gas and memory ops are OUTSIDE the fragment — a documented under-approximation matching the obligation surface the prior phases mechanized. Field bounds (app < 2³², slot refinement for read/write, a1, a2 < 2⁶⁴) are explicit refuses (invalid-execbounds).
- **Order-preserving canonical encoding.** ``TraceBytes = 0x03 || n4 || (op1 || app4 || a18 || a28)*`` in EXACT trace order — no sorting, no deduplication, no set semantics anywhere; two traces with the same multiset in different orders have different bytes and digests by construction (mechanized: EX3c/EX3d).
- **Execution identity.** ``ExecutionId = H(0x1F || parentStateRoot || PayloadDigest || ConfigDigest || EnvDigest)`` with domain-separated component digests (0x11 payload, 0x12 env, 0x13 config). Only fields with a demonstrated security role are included; the parent state root is the Phase 1E anchored root of the parent record, making `PreRoot + ExecutionId + TrueEffects + PostRoot` the binding chain.
- **Witness.** Eight fields (version, parentRoot, executionId, payloadD, declaredD, traceD, postRoot, configD), canon 0x57..., digest checked by the gate with full coherence (each witness field pinned to the record or the recomputed context). **Signature ≠ authenticity** is explicit.
- **Trace semantics.** ExecTraceWrites (last-write-wins), ExecReplayMap / ExecReads0k / ExecReplay0k (the verifier-side replay validating every read observation against the working state and the exact final composition), ExecProj (π_P — reads dropped, order preserved), and ExecDeclared0k — the Level-3 core equality $H(\text{canon}(\text{declared})) == H(\text{canon}(\text{Writes}(\text{trace})))$.
- **Not an Ethereum trace format.** No RLP, no EIP-7685, no consensus structure is claimed; the model is minimal and fully mechanizable.

4. The additive record and gate

LinRecF = (LinRecE, execId, effTraceDigest, execWDigest, execConfigDigest)

with CanonCoreF = CanonCoreE || (new fields) — CanonCoreE is a strict PREFIX; the base childE is carried unchanged and the legacy chains run unchanged on the projections. VerifyLineageF is a first-fail chain over the unchanged Phase 1E chain, adding the six new verdicts above and valid-f. The full accept condition is **14 atomic conjuncts** (legacy

chain $\wedge$ corrected-composition $\wedge$ capacity $\wedge$ padkey $\wedge$ state-root $\wedge$ anchor $\wedge$ childE $\wedge$ execbounds $\wedge$ execonfig $\wedge$ execid $\wedge$ efftrace $\wedge$ effdecl $\wedge$ witness $\wedge$ effbind); the task brief's 12-candidate list is recorded as superseded by the actual decomposition — not silently reconciled.

5. Mechanized proofs (Haskell backend, kprove)

phase1f/proofs/ — 26 claims, one per file (part3_k_proofs.txt):

Group	Claims	Outcome
Identity + canonical trace exactness	EX1, EX2a-c, EX3b-d	7 PROVED
Trace replay semantics	EX4a-c	3 PROVED
Record binding (prefix, childE invariance, childF, witness digest)	EX5a-c, EX5e	4 PROVED
Branch-wise gate certificate + accept direction	EXB1-EXB6 (+EXB4b), EXBACC	8 PROVED
Single-implication form	EXGATE	NOT PROVED at the simplification budget (disclosed; 1E GA1 precedent; content carried branch-wise)
Negative controls (designed fail)	NC1 (stuck — the A-F1 hash-injectivity boundary), NC2, NC3	fail as designed

Total: 22 PROVED. Disclosed not-mechanized attempts (kept verbatim in proofs/not_mechanized/ with reasons): EX3a (symbolic order-preservation induction), EX4d (cross-slot map commutativity), EX5w (the replay-writes lemma — the gate does not depend on it), EX5d/f/g/h (honest-construction coherence over fully symbolic maps — the AuthRootOf path; content pinned concretely by the LLVM demo + the cross-layer bytes). Non-vacuity: the accept premise is satisfiable — valid-f is a permanent demo case on the LLVM backend with real keccak + secp256k1, mirrored by the Python suite; NC3 (the vacuity probe) fails as designed.

6. The abstract demonstration suite (LLVM, real krypto)

srw3exec-demo.k — the Oracle $\rightarrow$ Lending $\rightarrow$ Liquidator case study as a three-record LinRecF chain with execution-identity and parent-root threading (part4_k_demo_suite.txt; the concatenated suite exceeds the 3 GB sandbox as ONE term, so the SAME cases run as partitioned programs — none skipped, documented):

Case	Construction	Verdict
pos0/1/2	honest chain (oracle update; borrow reading the price; liquidation with an intermediate same-slot write corrected by the final write)	valid-f $\times$ 3
f8a/f8b/f8 diff	equal final state, divergent traces (trailing redundant read): BOTH accepted, digests differ	valid-f; trace-digests differ

Case	Construction	Verdict
neg1 (F1)	hidden write in the trace, declared unchanged	invalid-effdecl
neg2 (F2)	read suppressed from the presented trace, digest covers the full trace	invalid-efftrace
neg3 (F4)	same-slot reorder with changed value, fully re-forged presentation	invalid-anchor (Level 2 catches first — recorded at its actual layer)
neg3x (F4 boundary)	cross-slot reorder, same final map, re-bound trace	valid-f BY DESIGN (fragment: cross-slot order is not security-relevant; the trace digest still binds it — neg3xdiff=true)
neg4 (F5)	stale parent-root replay	invalid-execid
neg5 (F6)	fork-spec replacement	invalid-execonfig
neg6 (F7)	witness bound to a foreign trace digest	invalid-witness
neg7	bounds refuse (app = 2^{32})	invalid-execbounds
neg8 (F2 residual)	FALSE READ OBSERVATION, every digest coherently re-bound, all legacy+L2 green	invalid-effbind

F8's demonstration is the standing discipline made concrete: the model never infers trace equality from post-state equality nor the converse.

7. Python mirror and mutation harness

phase1f/python/ mirrors the K semantics exactly (exec_model.py — canon, identity, witness, replay, gate) and re-runs the full case table (test_exec_verify.py: **16/16** verdict cases + 3/3 byte checks against the K values) plus a hostile-mutation harness (exec_mutations.py: **15/15** as expected — the producer re-signs and re-digests everything forgeable; rejections at the documented first-fail layers, including m01-m02 digest flips, m03-m06 witness mutations, m07-m09 declared/trace tampering, m10-m12 context substitutions, m13 stale anchor; b01/b02 are DOCUMENTED BOUNDARIES: the cross-slot order swap is accepted by design, and the fully consistent hidden-write forgery is caught by the Level-2 anchor — the A-F3 boundary at the abstract layer, closed at the KEVM layer).

8. The instrumented-execution layer (KEVM) — the Level-3 closure demonstration

k/kevm/srw3-exec-evm.k (+ demos; part5_kevm_binding.txt):

- **Execution-derived traces.** Additive #next [0P] interception rules (priority 20, the frozen loop untouched) log the four fragment opcodes into a RESERVED TRACE ACCOUNT (4096, outside the observed state) with a (last-run, last-pc) guard so multi-run traces record exactly one event per execution step, in actual execution order, with the actual operands. The trace never passes through producer hands.
- #w3GateF(anchor, payload, declared, presented). The record binds the

PRESENTED artifacts (abstract-layer discipline); VerifyLineageF runs on the presented trace; on top of valid-f, the LEVEL-3 EXECUTION CHECK confronts the declared effects with the write projection of the EXECUTION-DERIVED trace.

- **Demos** (all with real keccak + secp256k1, obligations green, commit/lineage atomicity preserved — rejects restore all storages and append no lineage record):

- evm_exec_commit — honest: **valid-f** → **COMMIT**;

- evm_exec_hidden — the state-invisible hidden write (SSTORE of a slot's own value; L1, L2 and every abstract layer are BLIND to it; the presentation is fully coherent): ****invalid-declared-exec** → **REJECT** + **RESTORE**** — the execution-derived evidence is the **ONLY** layer that sees it; this is the phase's headline demonstration;

- evm_exec_tamper — tampered presented trace: ****invalid-effdecl** → **REJECT** + **RESTORE****.

- Multi-gate executability required additive carrier/run hoisting rules (1E recorded single-gate-per-run only) — disclosed as new additive rules, frozen files untouched.

9. Cross-layer byte consistency

scripts/phaself/part6_crosslayer.py (part6_crosslayer_bytes.txt):

9/9 byte-equality checks MATCH across Python ↔ K-abstract (llvm, real krypto) ↔ KEVM instrumented run: trace canon bytes, execlD, childF (abstract case), and for the KEVM commit record: config digest, execution id, trace digest, witness digest, the anchored state root (embedded from the Python reference — acceptance certifies root equality by construction), and the executed-trace canon bytes decoded from the tracer storage of the real run.

10. Countermodels CM-F1..F9 (all demonstrated; expectations recorded)

I D	Construction	Rejected at
F 1	hidden write	invalid-effdecl (abstract); invalid-declared-exec when state-invisible (KEVM)
F 2	suppressed read	invalid-efftrace (suppression); false observation → invalid-effbind (neg8)
F 3	declared-but-never-executed effect	invalid-effdecl (declared ≠ Writes(trace))

I D	Construction	Rejected at
F 4	fake order	same-slot value change → invalid-anchor (L2; the honest anchor pins the post); cross-slot → accepted BY DESIGN with trace-digest distinctness (fragment boundary)
F 5	context replacement (stale parent root)	invalid-execid
F 6	fork-spec replacement	invalid-execonfig
F 7	producer-signed fake effects (fully coherent forgery)	abstract layer: the binding holds by construction (the honest disclosure — A-F3); the same construction is REJECTED where the trace source is execution-derived (KEVM hidden demo); anchor-visible variants are caught at L2
F 8	equal final state, trace divergence	NOT an attack: both records accepted, digests distinct — the model does not assume same-state ⇒ same-trace (nor the converse)
F 9	hidden cross-application interaction	the interaction's write is an effect: declared ≠ Writes(trace) → invalid-effdecl; if also state-hidden → invalid-declared-exec (KEVM) / L2 anchor for state-visible variants

11. Prior-art positioning (targeted; `part7_prior_art.txt`)

Technique	The distinction
EIP-8025 (execution/state witnesses over RPC)	authenticates STATE pieces to clients from a serving node; no K semantics, no obligation gate, no lineage threading, no ordered effect trace
zkEVM provers	prove GENERIC execution validity — the frozen standing distinction (generic validity ≠ SRW3 security validity) applies verbatim; the obligation surface, lineage, and anchoring are absent; complementary as a future witness SOURCE
Stateless-Ethereum / Verkle witness formats	authenticate the state TOUCHED, not the ordered effect stream; no execution identity, no read-observation binding (the oracle-staleness class), no lineage
Attested execution receipts (TEE/quorum)	trust replaces verification; SRW3 recomputes everything mechanized and states signature ≠ authenticity explicitly
Fraud-proof / evidence systems	retroactive and challenge-gated; SRW3's gate is commit-time with lineage atomicity
KEVM contract verification	proves properties of programs; 1F proves properties of a GATE over records/traces/identities/anchors — a different artifact class on the same engine

No technique surveyed combines: order-preserving digest-bound effect traces, a domain-separated execution identity over (parent root, payload, config, env), witness-coherent lineage records, replay-validated read observations, and a commit-time security-obligation gate over real EVM execution. The comparison upgrades no evidence tier.

PHASE 1F RESULTS

Research question: can the gate bind its security decision to authenticated evidence of execution-derived effects (PresentedEffects == TrueEffects-(execution), Level 3)?

Status: ANSWERED – qualified positive at Level 3 as a BINDING result under stated assumptions; three-level separation preserved; the authenticity boundary stated, not assumed away

Effect-trace fragment: Read/Write/Call/Return over SLOAD/SSTORE/CALL/RETURN; order-preserving canonical encoding (no sorting/dedup/set semantics); explicit bounds refuses; NOT an Ethereum trace format

Execution identity: $H(0 \times 1F \parallel \text{parentRoot} \parallel \text{payloadD} \parallel \text{configD} \parallel \text{envD})$ – domain-separated components; parent root = the 1E anchored root (PreRoot + ExecutionId + TrueEffects + PostRoot chain)

Witness: 8-field canon + full gate coherence; signature $\neq$ authenticity explicit

Record: LinRecF additive; CanonCoreE strict prefix of CanonCoreF; childE carried unchanged

Gate_F: first-fail; 14 atomic conjuncts (the brief's 12-candidate list recorded as superseded); new verdicts: invalid-execbounds/execconfig/excid/efftrace/effdecl/witness/effbind; effdecl = the LEVEL-3 CORE CHECK (declared effects == Writes(trace), decidable, not post-relative)

Mechanized: 22 kprove claims PROVED (identity/canon exactness, replay, record binding, branch-wise EXEC-EFFECT-BINDING certificate + accept direction); EXGATE single form BLOCKED (budget, disclosed); NC1-3 designed-fail; 7 disclosed not-mechanized attempts (kept)

Demonstrated (abstract): 3-record Oracle->Lending->Liquidator chain (valid-f x3), F1-F9 classes, F8 trace-sensitivity, cross-slot order boundary – LLVM, real crypto, partitioned programs

Demonstrated (KEVM): EXECUTION-DERIVED traces via additive #next-interception; honest -> valid-f COMMIT; state-invisible hidden write -> invalid-declared-exec REJECT+RESTORE (the headline: only the execution evidence sees it); tampered trace -> invalid-effdecl REJECT

Cross-layer bytes: 9/9 MATCH (P <-> K-abstract <-> KEVM)

Python mirror: 16/16 cases + 15/15 mutations (2 documented boundaries)

Zero regression: frozen 1D/1E surface untouched; baseline re-verified at start (L7 10/10, mutations 21/21, comparisons 33/33, 1E suites, K composition suite)

Assumptions: A-F1 keccak collision resistance (REQUIRES CRYPTOGRAPHIC ASSUMPTION); A-F2 structural injectivity; A-F3 trace authenticity = verifier-domain boundary at the abstract layer, CLOSED by construction at the KEVM layer, else REQUIRES CLIENT/PROTOCOL SUPPORT (re-execution); A-F4 explicit bounds refuses; A-F5 parent-root threading (inherits A-E3)

Level 1 (Phase 1D-R1): unchanged – presented-record internal self-consistency (frozen surface, zero regression)

Level 2 (Phase 1E): unchanged – anchored state commitment

(Merkle root == anchor); anchor provenance still **REQUIRES CLIENT/PROTOCOL SUPPORT**
 Level 3 (Phase 1F): binding **PROVED BY K** (branch-wise certificate + accept direction); execution-grounded closure **DEMONSTRATED BY KEVM** (instrumented real execution); general historical-truth authentication **REQUIRES CLIENT/PROTOCOL SUPPORT** (re-execution) – recorded, not silently closed
 Anti-vacuity: pos=valid-f permanent demos (abstract + KEVM) + Python E0-style witness; NC3 fails as designed
 Honest disclosures: EXGATE single-form **BLOCKED**; 7 not-mechanized attempts kept; sandbox-reset recovery recorded; 3 GB sandbox memory constraint documented (partitioned demo programs); the abstract-layer consistent-forgery boundary (b02) stated as the A-F3 boundary, closed at the KEVM layer only by demonstration
 Git: branch phase1f derived from phasele closure 8b26cb74 (scientific anchor 0d826c0f); no history rewrite of any milestone branch
 Artifact bundle: SRW3-Phase1F-Artifact-Bundle.zip + .sha256 sidecar (MANIFEST 422 entries, round-trip ALL OK; cumulative 0-D..1F)
 Completion verdict: **COMPLETE** – all completion criteria evaluated below; no criterion silently waived

12. Completion criteria evaluation

#	Criterion	Verdict
1	TrueEffects vocabulary + documented fragment	SATISFIED (MODEL.md §2 frozen; bounds explicit)
2	Execution identity binding (parent root, payload, config, env)	SATISFIED (recomputation layer; F5/F6 negative controls)
3	Canonical, deterministic, independently recomputable EffectDigest	SATISFIED (order-preserving canon; 9/9 cross-layer bytes)
4	ExecWitness + coherence; signature ≠ authenticity	SATISFIED (8-field witness; gate layer L21)
5	Non-trivial theorem (strongest provable form)	SATISFIED branch-wise (8 gate claims + accept direction); single form BLOCKED and disclosed
6	Negative controls with expected verdicts	SATISFIED (NC1-3 + CM-F1..F9 + 15 mutations)
7	Level-2 preserved	SATISFIED (frozen diff empty; baselines green)
8	Three-level separation preserved	SATISFIED (§11 verdicts stated per level; no merging)
9	Prior-art comparison incl. the standing distinction	SATISFIED (part7; scoped, tier-neutral)
10	Independently reproducible	SATISFIED (transcripts + pinned recipes + per-claim raw outputs)
11	Honest witness accepted / forgery rejected / fake effects fail	SATISFIED (commit/hidden/tamper demos; b02 boundary disclosed and closed at KEVM)
12	Packaging: report, bundle, provenance, git	SATISFIED by this closure

13. What is proved, demonstrated, and open

- **PROVED BY K (22 claims).** Identity and canonical-trace exactness; the replay semantics; the record binding (prefix + childE invariance + childF + witness digest); the branch-wise **EXEC-EFFECT-BINDING** certificate (every extension verdict characterized exactly) and the accept direction

(all-14 $\implies$ valid-f). No claim consumes hash evaluation; injectivity stays pinned to A-E1/A-F1 (NC1 stuck by design).

- **DEMONSTRATED BY KEVM.** The execution-derived trace construction over real opcode execution; honest $\rightarrow$ valid-f COMMIT; the state-invisible hidden write $\rightarrow$ invalid-declared-exec REJECT+RESTORE; the tampered presentation $\rightarrow$ invalid-effdecl REJECT+RESTORE; 9/9 cross-layer bytes.

- **NOT YET MECHANIZED (disclosed).** EXGATE's single-implication form (content carried branch-wise); the replay-writes lemma (the gate does not depend on it); cross-slot map commutativity; symbolic-universe coherence pins (concretely pinned by demos + bytes).

- **REQUIRES CLIENT/PROTOCOL SUPPORT.** Trace authenticity for abstract-layer records (re-execution or an attested execution-derived source); anchor provenance (inherited from 1E, untouched — no circular strengthening).

- **Standing open items (unchanged).** Tier-3 universal reconciliation; the general cryptographic theorem form; the result remains a concrete demonstration within the modeled verifier domain.

Phase 1G remains out of scope and is not entered by this closure.

14. Artifact index

Artifact	Path
Model spec (FROZEN)	phase1f/semantics/MODEL.md
K semantics (additive)	phase1f/semantics/srw3exec.k + demo srw3exec-demo.k
K proofs + negative controls	phase1f/proofs/*.k, proofs/not_mechanized/
Python reference + suites	phase1f/python/{exec_model,test_exec_verify,exec_mutations}.py
KEVM binding + demos	k/kevm/srw3-exec-evm.k, srw3-exec-evm-demos.k, k/kevm/demos/evm_exec_*.srw3evm
Transcripts (Parts 0-7)	phase1f/transcripts/ (+ part3_per_claim/ raw outputs)
Cross-layer + prior-art scripts	scripts/phase1f/
Report	phase1f/report/SRW3-Phase1F-Execution-Binding-Report.{md,pdf}
Provenance	phase1f/provenance/PROVENANCE.md